Maya LeBlanc

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Technical Skills

Design/CAD: SolidWorks, FEA, PDM, AutoCAD, Inventor, Onshape, Revit, Autodesk 3ds Max
Robotics/Hardware: Raspberry Pi, ESP32, Arduino, 3D printing, machining
Programming/Tools: Python, Bash, MATLAB, ROS 2 (Jazzy), Gazebo Harmonic, RViz, URDF/Xacro, ros2_control, ros_gz_bridge, Open3D, NumPy/SciPy, Git, Linux (Ubuntu 24.04)
Data/Research: Excel, Minitab Statistical Software, NCBI GEO
3D Perception/Algorithms: Point cloud processing, RANSAC and geometric fitting, DBSCAN segmentation, Poisson surface reconstruction, depth-camera simulation, grasp planning
Other: LaTeX, Microsoft 365, Photoshop, Illustrator, Ansys Granta EduPack

Projects (Selected)

ROS2 Robot Simulation and Sensor Integration — May 2026 – Current
- Built and simulated a differential-drive robot in ROS2; modeled geometry in URDF/Xacro, integrated LiDAR via TF transforms, and enabled keyboard teleoperation in Gazebo.

EEG-Controlled Prosthetic Hand (Project AURA) — May 2025 – Jun 2025
- Designed EEG-driven prosthetic with Arduino, Python signal processing, and 3D-printed ergonomic frame.

Experience (Selected)

Communications Systems Lead, Mechanical Subteam — August 2026 – Current
McMaster Mars Rover Team — *Hamilton, ON*
- Led end-to-end design and deployment of the ground basestation, antenna, and pan-tilt camera assembly, taking the communications subsystem from concept to field operation.
- Produced engineering drawings and machined structural components, conducting FEA to validate mechanical integrity under rover operating loads.
- Promoted from sole contributor to sub-team lead after independently delivering the subsystem; now direct a team of 2, owning technical decisions, integration, mentorship, and sub-team budgets and timelines.

Undergraduate Robotics Researcher — Aug 2025 – Current
McMaster Robotics and Manufacturing Automation Laboratory — *Hamilton, ON*
- Developed a novel grasp-planning algorithm for agricultural robotics in MATLAB, using multi-height cross-sectional slicing and fruit-specific deformability modeling to generate grasps for 5 fruit geometries from 3D mesh input.
- Built an end-to-end validation pipeline in ROS 2 (Jazzy) and Gazebo Harmonic to test the algorithm on simulated depth-sensor data, integrating a 3-fingered depth-camera gripper for automated point-cloud capture, segmentation, and shape completion across dozens of programmatic viewpoints.
- Validated the perception-to-grasp system over 108 trials spanning 3 sensor-noise levels, achieving $<$0.5 mm diameter-estimation bias and $<$2.5 mm MAE at realistic noise, with an automated quality gate filtering $\sim$30% of degenerate viewpoints.
- Bridged a CAD-mesh-based planning algorithm to realistic sensor input by engineering a partial-view-to-mesh reconstruction pipeline (depth capture → segmentation → sphere fitting → symmetry completion → watertight meshing), enabling the planner to operate on noisy, occluded point clouds instead of idealized geometry.

Undergraduate Researcher — Aug 2025 – Current
McMaster Health Sciences Lupus Clinic and Aletheia — *Hamilton, ON*
- Founded a computational immunogenetics initiative combining gene expression analysis and ML to develop diagnostic biomarkers for autoimmune disease, in collaboration with doctors at McMaster, UofT, and McGill.

Awards (Selected)

ISEF 4th Place Grand Award in the Biomedical and Health Sciences Cateory (2024)

CWSF 4 Awards (2 Bronze, 1 Merit, 1 Intact Climate Change Resilience Award) (2021-2024)

BASEF Best in Fair & Best Intermediate/Senior Project (2024)

BASEF 20 Awards (2019-2024)

Publications

LeBlanc, M. ”Analysis of CD16 +/-Monocytes and CD4 + T Cells to Identify Novel Gene Signatures and Develop a Diagnostic Tool for SLE.” *International Journal of High School Research,* 2025

LeBlanc, M. “Carbon Capture Depletion Using Catalyzed Photosynthesis to Promote Crop Growth.” *Canadian Science Fair Journal*, 2023.

Certifications (Selected)

Robotics Dev Fundamentals: Linux, C++, ROS2, micro-ROS — Robotisim (2025)

SOLIDWORKS Essentials Training — FEA Training Consulting Inc. (2025)

Education

McMaster University — Hamilton, ON
Candidate for BEng in Materials Engineering — *Sep 2025 – Apr 2029*

University of Waterloo — Waterloo, ON
Quantum School for Young Students, Institute of Quantum Computing — *Jul 2025 – Aug 2025*